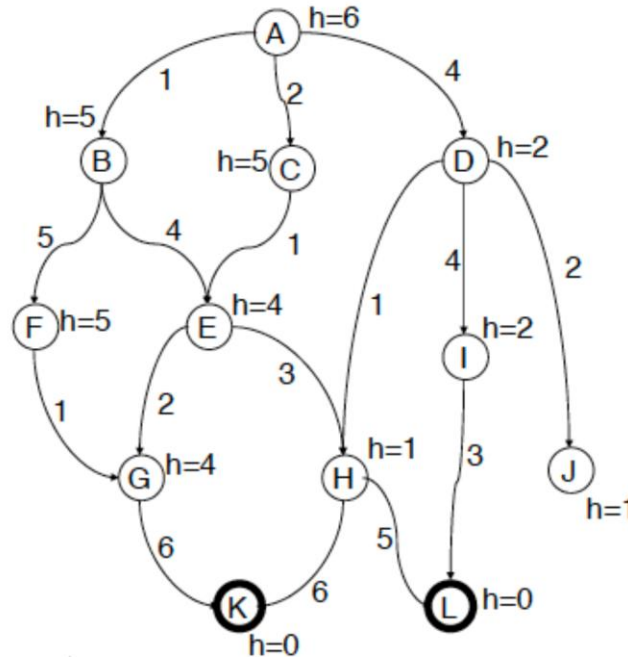


Informed Search

Lab 3

Exercise 1 (greedy best first, from A to the goal)

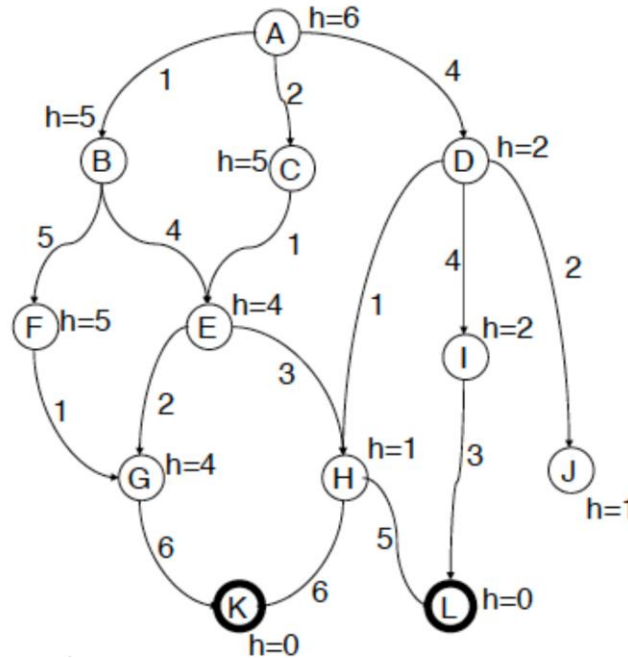


Copy the program from last class (Lab2) and modify it to implement the greedy best first search algorithm (GBFS) and apply it to this graph. Play with the heuristics and see what different heuristic functions would yield as solutions. (Don't modify the path costs).

Today's exercises are quite a bit more difficult than the first two labs. Ask questions when you need to and don't be discouraged.

The goal states are K and L.

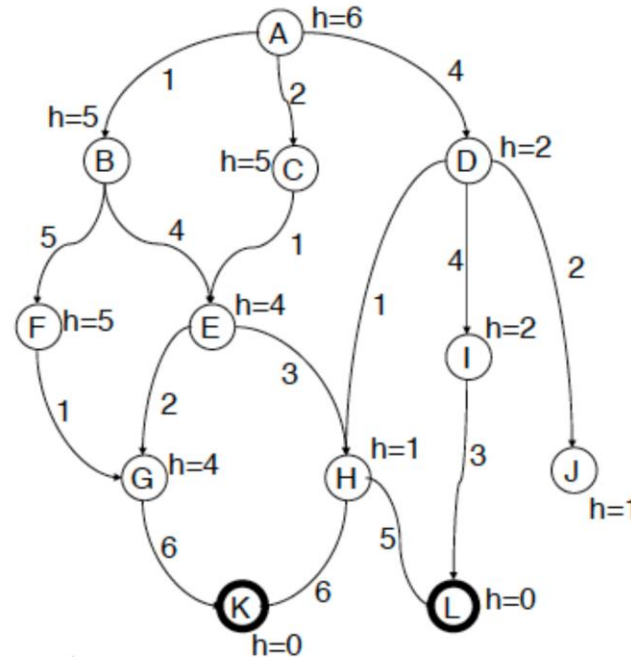
Exercise 2 (A*, from A to the goal)



Copy the program from exercise 1 and modify it to implement A* search for this graph. Compare the solution path to GBFS. Play with the heuristics and see what different heuristic functions would yield as solutions.

The goal states are K and L.

Exercise 3 (Weighted A*, from A to the goal)



Modify the program from exercise 2 to implement weighted A*. Compare the solution paths to the previous exercises. Play with the heuristics and see what different heuristic functions would yield as solutions. Play with different weights and compare paths and number of nodes explored.

The goal states are K and L.

Homework (A* Vacuum Cleaner)

You should be able to use much of the code from the exercise. You will need to make a few changes though, especially with how to handle the state.

1. The state of the search should be represented with three elements: a state, a path and a cost. Ultimately, cost is defined as the number of moves taken to achieve the goal state from the initial state.
2. While moves and paths resemble those in lab 1, solutions (output) are the path, the cost of the solution and the number of explored nodes.

Challenge: Can you generalize it?